

Climate

i. Monsoon

*The word '**Monsoon**' is derived from the Arabic word '**Mausim**' which means season. *Traditionally, the 'Monsoon' is related to the change in the pattern of monsoonal winds through which rain occurs in South and Southeastern Asian countries. India receives near-about **75%** of its total precipitation from the southwest monsoon. The monsoon is active from **June to September**, which is very important for agriculture in India. *The monsoon arrives at the southern tip (Kerala) of the Indian peninsula by the first week of June. India's climate has characteristics of both tropical and subtropical climates.



Origin of Indian Monsoon-

Monsoon is a result of **differential rates of heating and cooling of land and sea**. The sun is vertical over the Tropic of Cancer in summer season, and the northern regions of the Indian landmass get heated. This leads to the formation of low pressure conditions over the Indian subcontinent. On the other hand, the Indian Ocean is colder, so high pressure is generated, leading to northward shifting of Inter Tropical Convergence Zone. ***The Inter Tropical Convergence Zone (ITCZ)** is a low-pressure belt near the equator where trade winds of the northern and the southern hemispheres converge. *Due to the northward shift of the ITCZ, the trade winds of the southern hemisphere cross the equator between **40°** and

60° E longitudes and starts blowing from the southwest to the northeast under the influence of the **Coriolis** force. It becomes southwest monsoon. *In winter, the ITCZ moves southward, and so the reversal of winds from northeast to south and southwest takes place. They are called **northeast monsoons**. *The sudden onset of the moisture-laden winds associated with violent thunder and lightning is often termed as the '**burst**' of the monsoon. The monsoon may burst in the first week of June in the coastal area of Kerala.

*The monsoon approaches the Indian landmass in two main branches:

- (i) The Arabian Sea branch
- (ii) The Bay of Bengal branch

*The Tibetan Plateau has a very broad impact on the Indian climatic pattern.

*Rainfall received from the southwest monsoons is seasonal in character, occurring between June and September. Monsoonal rainfall is largely governed by relief or topography. The monsoon rainfall has a declining trend with increasing distance from the sea. The rainless intervals are known as '**breaks**' in the monsoon.

The Retreat of the Monsoon - The months of October and November are known for **retreating monsoons**.

*The weather in the retreating monsoon is dry in north India whereas it is associated with rain in the eastern part of the Peninsula. The widespread rain in this season is associated with the passage of cyclonic depressions that originate over the Andaman Sea and manage to cross the eastern coast of the Southern Peninsula. These tropical cyclones are very destructive. A bulk of the rainfall on the Coromandel coast is derived from these depressions and cyclones. Such cyclonic storms are less frequent in the Arabian Sea.

Sr. No.	Local Storms of Hot weather Season	Region
(i)	Mango Shower	Kerala, and Coastal areas of Karnataka
(ii)	Blossom Shower	Kerala
(iii)	Norwesters (Kalbaisakhi)	West Bengal and Assam
(iv)	Loo	Northern Plains

***El-Nino** is a complex weather system bringing drought, floods and other weather extremes in different parts of the world. It occurs with the sudden appearance of warm currents off the coast of Peru (warm equatorial current replaces cold Peruvian current).

1. The term 'Monsoon' is originated from :

- (a) Arabic Language (b) Spanish
(c) Hindi (d) English

M.P.P.C.S. (Pre) 2013

Ans. (a)

The word 'Monsoon' has been derived from the Arabic word 'mausim' which means 'season'. Traditionally, Monsoon is conceived as a seasonal reversal of winds because of which rainfall occurs in South and Southeast Asia.

2. Assertion(A) : India is a Monsoonal Country.

Reason (R) : The high Himalayas impart it climatic distinctiveness.

Code :

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

U.P.P.C.S. (Pre) 1996, 1998

Ans. (a)

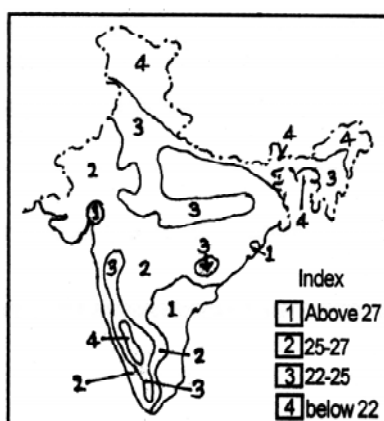
U.P. Lower Sub. (Pre) 1998

The climate of India is essentially monsoonal. Almost all the features of the monsoon climate are found in India. So, the assertion (A) is correct.

The lofty Himalayas in the north along with their extension act as an effective climatic divide. The towering mountain chain provides an invincible shield to protect the subcontinent from the cold northern winds. The Himalayas also trap the monsoon winds, forcing them to shed their moisture within the subcontinent.

So, reason (R) is the correct explanation of (A). Hence, option (a) is the correct answer.

3. Consider the map of India given below. The map read with index provided refers to the distribution of.



- (a) Winter Rainfall (b) Pressure
(c) Annual rainy days (d) Mean Temperature

I.A.S. (Pre) 1994

Ans. (d)

India is home to diverse climates. It hosts humid tropical to temperate zones. Thus, different geographical regions experience different temperatures. The above map depicts annual mean temperature. Northern India is relatively warmer than the south. Similarly, during winter, the northern area is colder than the southern area.

4. Assertion (A) : India has a tropical Monsoon type climate.

Reason (R) : India is located exactly between the tropical latitudes.

Code :

- (a) Both (A) and (R) are true, and (R) is the Correct explanation of (A).
(b) (A) is true, but (R) is false
(c) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(d) (A) is false, but (R) is true.

Jharkhand P.C.S. (Pre) 2013

Ans. (b)

The climate of India is tropical monsoonal and it is situated between 8°4' to 37°6' north latitudes. It is not situated between tropical latitudes i.e, 30° north to 30° south latitudes. So, (A) is correct and (R) is false.

5. In which of the following States does the Monsoon arrive first?

- (a) Assam (b) West Bengal
(c) Maharashtra (d) Kerala

U.P.P.C.S. (Pre) 2006

Ans. (d)

In India, monsoon arrives first in Kerala. India receives 75 percent of its total rainfall from south-western monsoon which starts from Kerala and is active from June to September every year.

6. The general direction of flow of summer monsoon in India is –

- (a) from South to North
(b) from South West to South East
(c) from South East to South West
(d) from South West to North East

M.P.P.C.S. (Pre) 2012

Uttarakhand P.C.S. (Pre) 2006

Ans. (d)

In summer, the monsoon flows from South-West to North-East direction. During winter while retreating back, it flows from North-East to South-West direction. Maximum rainfall in India occurs due to south-west monsoon.

7. **Monsoon originates by which of the following winds in India?**

- (a) South-West winds
- (b) South-East winds
- (c) North-East winds
- (d) North-West winds

U.P. R.O./A.R.O. (Mains) 2016

Ans. (a)

See the explanation of above question.

8. **High temperature and low pressure over the Indian Subcontinent during the summer season draw air from the Indian Ocean leading to the in-blowing of the :**

- (a) South-East monsoon
- (b) South-West monsoon
- (c) Trade winds
- (d) Westerlies

I.A.S. (Pre) 1996

Ans. (b)

The south-west monsoon is responsible for most of the rainfall in India. It is caused by high temperature and low pressure over the Indian subcontinent during the summer season which draws air from the Indian Ocean.

9. **Which one of the following seas has highest salinity?**

- (a) Red Sea
- (b) North Sea
- (c) Arabian Sea
- (d) Mediterranean Sea

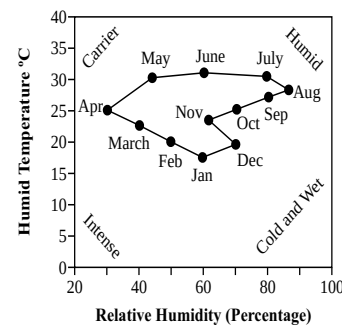
Uttarakhand P.C.S. (Pre) 2024

Ans. (a)

The Salinity for normal open ocean ranges between 33‰ and 37‰. In the landlocked Red Sea, it is as high as 41‰, while in the estuaries and the Arctic, the salinity fluctuates from 0-35‰, seasonally. In hot and dry regions, where evaporation is high the salinity sometimes reaches 70‰.

The North Sea, in spite of its location in higher latitudes, records higher salinity due to more saline water brought by the North Atlantic Drift. The Baltic Sea records low salinity due to influx of river water in large quantities. The Mediterranean sea records higher salinity due to high evaporation.

10. **Consider the climate diagram given below:**



The above diagram is related to the climate of which of the following regions?

- (a) North –West region of India.
- (b) South region of India
- (c) Central region of India
- (d) North-East region of India

I.A.S. (Pre) 1993

Ans. (d)

The above map depicts the climatic condition of North-East region of India.

11. **Which one of the following is the driest place?**

- (a) Mumbai
- (b) Delhi
- (c) Leh
- (d) Bengaluru

56th to 59th B.P.S.C. (Pre) 2015

Ans. (c)

Among the places given in the options, Leh receives the least rainfall. It is the driest place among the four.

12. **The January isotherm taken as a basis for dividing India into tropical and subtropical zones is :**

- (a) 21°C
- (b) 18°C
- (c) 12°C
- (d) 15°C

I.A.S. (Pre) 1997

Ans. (b)

The January isotherm taken as the basis for dividing India into tropical and subtropical zones is 18°C.

13.



In the shaded area of the above map, the mean temperature for the month of July varies between :

- (a) 22.5°C-25.0°C
- (b) 25.0°C-27.5°C
- (c) 27.5°C-30.0°C
- (d) 30.0°C-32.5°C

I.A.S. (Pre) 2001

Ans. (b)

In the shaded area of the given map, the mean temperature for the month of July varies between 25°C to 27.5°C.

14. In which part of India is the daily range of temperature highest?

- (a) Eastern coastal areas.
- (b) Interior areas of Chhattisgarh plain
- (c) Andaman island
- (d) Desert areas of Rajasthan.

Chhattisgarh P.C.S. (Pre) 2011

Ans. (d)

Desert areas of Rajasthan have the highest daily range of temperature. Daily temperature variation is one of the main features of desert climates. Deserts are hot during the day but cold during the night.

15. Which are the months for monsoon usually in Tamil Nadu?

- (a) March-April
- (b) June-July
- (c) September-October
- (d) November-December

M.P.P.C.S. (Pre) 2005

Ans. (d)

The period of October to December is referred to as North East Monsoon season over Peninsular India. It is the period of rainfall over South India, particularly in Andhra Pradesh, Rayalaseema and Tamil Nadu. This is the main rainy season in Tamil Nadu region. During this monsoon on 26 December, 2004, Tamil Nadu was struck by a Tsunami. So, the answer is (d).

16. The Indian monsoon is indicated by seasonal displacement because of

- (a) Differential temperature of land and sea
- (b) Cold wind of middle Asia
- (c) Excess similarity of temperature
- (d) None of the above

39th B.P.S.C. (Pre) 1994

Ans. (a)

The Indian monsoon is indicated by seasonal displacement because of the differential temperature of land and sea. Some of the important concepts about the origin of the Monsoon are:

(1) Thermal concept: It propounds that the primary arrival of the annual cycle of the Indian monsoon circulation is the differential heating effect of the land and the Sea.

(2) Dynamic concept: Put forward by Flohn, according to this concept, monsoon is the result of seasonal migration of planetary winds and pressure belts.

17. Consider the following statements :

- 1. The duration of the monsoon decreases from southern India to northern India.
- 2. The amount of annual rainfall in the northern plains of India decreases from east to west.

Which of the statement(s) given above is/are correct?

- (a) Only 1
- (b) Only 2
- (c) 1 and 2
- (d) Neither 1 nor 2

I.A.S. (Pre) 2012

Ans. (c)

The amount of rainfall decreases from east to west owing to the progressive decrease in the humidity of the winds. The southern part of India being near the sea gets rainfall early and for a longer duration than the northern part. Thus, both statements are correct.

18. With reference to 'Indian Ocean Dipole (IOD)' sometimes mentioned in the news while forecasting Indian monsoon, which of the following statements is/are correct?

- 1. IOD phenomenon is characterised by a difference in sea surface temperature between in sea surface temperature between tropical Western Indian Ocean and tropical Eastern Pacific Ocean.
- 2. An IOD phenomenon can influence an El Nino's impact on the monsoon.

Select the correct answer using the code given below:

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

I.A.S. (Pre) 2017

Ans. (b)

The Indian Ocean Dipole (IOD) is also known as the 'Indian Nino'. IOD is an irregular oscillation of sea-surface temperature in which the western Indian Ocean becomes alternately warmer and then colder than the eastern part of the Indian Ocean (not the tropical eastern Pacific Ocean). So, statement 1 is wrong. An IOD phenomenon can influence an El-Nino's impact on the monsoon. So, statement 2 is correct. Therefore, option (b) is correct.

19. Which of the following statements is not correct ?

- (a) El Nino is an ocean current, which appears off the Peru coast.
- (b) El Nino weakens the Indian Monsoon.
- (c) El Nino occurs every year.
- (d) La Nina strengthens the Indian Monsoon.

M.P.P.C.S. (Pre) 2024

Ans. (c)

El Nino appears as a warm oceanic current near the coast of Peru in the eastern and central tropical Pacific Ocean. El Nino generally has an adverse effect on India's Southwest Monsoon, often leading to drought like conditions. The El Nino phenomenon occurs irregularly at intervals of approximately two to seven years.

La Nina is the condition opposite of an El Nino. In a La Nina the tropical Pacific trade winds become very strong and an abnormal accumulation of cold water occurs in the central and eastern Pacific Ocean.

20. What are true in relation to jet stream?

1. Jet stream blow across Asian Continent at latitudes North of Himalayas parallel to Tibetan highlands.
2. Tibetan highland divides jet stream into North and South Branches.
3. The northern branch of jet stream plays important role in winter weather in India.

- (a) 1 and 2
- (b) 1 and 3
- (c) 1, 2 and 3
- (d) None of the above.

Chhattisgarh P.C.S. (Pre), 2022

Ans. (a)

During the winter, a distinct pattern of air circulation is observed higher up in the atmosphere. Significantly, westerly winds continue to affect western and central Asia extending approx. 9 to 13 km from west to east. At latitudes north of the Himalayas and almost parallel to the Tibetan highlands, these winds blow across the Asian continent. Thus the first statement is true.

Notably, these jet streams are obstructed by the hills of Tibet consequently, jet stream splits in two. One of its branches flows north of the Tibetan highlands and the other, which originates south of the Himalayas, blows eastward before merging again to form a single stream over China, Hence statement 2 is correct.

Notably, the northern branch of jet stream does not play an important role in winter weather in India. So statement (3) is false.

21. Given below are two statements, one is labeled as Assertion (A) and the other as Reason (R).

Assertion (A) : Agra and Darjeeling are located on the same latitude but temperature of January in Agra is 16° Centigrade whereas it is only 4° Centigrade in Darjeeling.

Reason (R) : Temperature decreases with height and due to thin air, places in the mountains are cooler than places in the plains.

Select the correct answer from the code given below :

Code :

- (a) (A) is true but (R) is false
- (b) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (c) (A) is false but (R) is true
- (d) Both (A) and (R) are true but (R) is not the correct explanation of (A)

U.P.P.C.S. (Pre) 2022

Ans. (b)

Under normal conditions, the temperature usually decreases with an increase in altitude in the troposphere at a rate of 1 degree for every 165 meters. Significantly, it is called the normal lapse rate. It is worth mentioning that latitudinal positions of Agra and Darjeeling are 27°17' and 27°2' respectively, which are almost same. Significantly, the temperature of January in Agra is 16° C whereas it is only 4° C in Darjeeling. So we can say that both (A) and (R) are true and (R) is the correct explanation of (A).

22. Amritsar and Shimla are almost on the same latitude, but their climate difference is due to-

- (a) The difference in their altitudes
- (b) Their distance from sea
- (c) Snowfall in Shimla
- (d) Pollution in Amritsar

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (a)

Amritsar (31°29'-32°03'N) and Shimla (30°45' - 31°44'N) are almost on the same latitude but their climate difference is due to the difference in their altitudes. Amritsar is located at approx. 756 feet, and Shimla is at approx. 7866.10 feet above the mean sea level.

23. Which one of the following types of climate is found in Chhattisgarh state ?

- (a) Humid South-East
- (b) Sub Humid Transitional

- (c) Sub Humid Littoral (d) Sub Humid Continental
Chhattisgarh P.C.S. (Pre) 2011

Ans. (a)

Humid South-East climate is found in the region of West Bengal, Chotanagpur, Odisha plain, southern Chhattisgarh, and the northeast part of Andhra Pradesh. So, option (a) is the correct answer. Sub-humid transitional weather extends to central Gangetic plains (eastern U.P. and western Bihar). Sub-humid littoral weather spreads in the Coromandel region and sub-humid continental weather spreads to upper Gangetic plains.

- 24. Which of the following regions is not affected much by the Arabian Sea branch of Monsoon?**

- (a) The Western Ghat (b) Deccan Plateau
(c) Madhya Pradesh (d) Chhattisgarh Basin
(e) None of these

Chhattisgarh P.C.S. (Pre) 2016

Ans. (d)

The Chhattisgarh Basin is not affected much by the Arabian Sea branch of the monsoon. Arabian Sea branch of the monsoon further splits into two branches after being obstructed by the Western Ghats. The first branch moves parallel to Aravalli range causing scanty rainfall and further mingles with the Bay of Bengal branch in North India. The second branch, after being obstructed by Vindhya mountains, causes rainfall in Gujarat, Madhya Pradesh and Western Chhattisgarh. Therefore, it is clear that the Chhattisgarh Basin is not much affected by the Arabian Sea branch of the monsoon.

- 25. Which of the following places experience a humid climate?**

1. Ahmedabad 2. Kochi
3. Ludhiana 4. Tejpur

**Select the correct answer from the code given below :
Code :**

- (a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 2 and 4 only

U.P. U.D.A./L.D.A. (Spl) (Pre) 2010

Ans. (d)

Kochi and Tejpur experience more humid climate due to heavy rainfall. In 2022, Kochi (Ernakulam) experienced rainfall of up to 3415.8mm, Ahmedabad- 564.0 mm, Ludhiana- 766.9mm and Tejpur – 1775.7 mm (Sonitpur). Compared to Ahmedabad and Ludhiana, Kochi and Tejpur experience more rainfall which makes their climate more humid.

- 26. The decline of monsoon is indicated by**

- (i) Clear Sky
(ii) Pressure condition in Bay of Bengal
(iii) Rising temperature on land

Select your answer using the following code –

- (a) only i (b) i and ii both
(c) i, ii and iii (d) ii and iii both

40th B.P.S.C. (Pre) 1995

Ans. (c)

When the monsoon returns from the north, there is a slight increase in temperature on land. The low pressure covers the Bay of Bengal and the sky is clear. Thus, all statements are correct.

- 27. Consider the following statements :**

- (1) The winds which blow between 30°N and 60°S latitudes throughout the year are known as westerlies.
(2) The moist air masses that cause winter rains in the North-Western region of India are part of westerlies.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2015

Ans. (b)

Westerlies flow between 30° and 60° latitudes of the northern and southern hemispheres. The statement mentions 30° N to 60°S which is wrong. The western cyclonic disturbances enter the Indian subcontinent from the west and northwest during the winter months, originate over the Mediterranean Sea and are brought into India by the westerly jet stream, causing winter rains in northern and northwestern India. They are a non-monsoonal precipitation pattern driven by westerlies.

- 28. Intertropical convergence zone (ITCZ), a low-pressure zone, is located:**

- (a) between trade winds and westerlies belt
(b) between westerlies and polar winds belt
(c) nearby the poles
(d) at the equator

M.P. P.C.S. (Pre) 2018

Ans. (d)

An Intertropical convergence zone (ITCZ) is an area of broad trough and low pressure in the equatorial latitudes. It is here that the Northeast and Southeast trade winds meet. This convergence zone is roughly parallel to the equator, but it shifts north or south along with the apparent motion of the sun. Hence, option (d) is the correct answer at the equator.

29. The dust storms originating in Rajasthan in May and June months are caused due to:

- (i) Origin of convectional currents at few places
- (ii) Aravalli hills are parallel to south-western winds
- (iii) Origin of fast speedy easterly winds

- (a) (i) and (iii)
- (b) (i), (ii) and (iii)
- (c) (i) and (ii)
- (d) Only (i)

R.A.S./R.T.S. (Pre) 2018

Ans. (d)

In the months of May and June, hot winds blow from the west, which is called Loo. Due to this heat, an area of low air pressure is formed here. To compensate for this low air pressure, air rises rapidly, bringing high dust storms with it. Hence, the origin of convectional currents at some places is responsible for the dust storms that occur in the May and June in Rajasthan.

30. Rajasthan receives very little rain because :

- (a) the monsoon fails to reach this area
- (b) it is too hot
- (c) there is no water available and thus, the winds remain dry
- (d) the winds do not come across any barriers to cause the necessary uplift to cool the air
- (e) None of the above / More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (d)

Rajasthan receives very little rainfall, as the winds do not cross any barriers, due to which they do not attain the required height to cool down. The Aravalli mountain ranges become a partial obstacle in the way, as they are aligned with the direction of the South-West monsoon winds. Hence, most of Rajasthan (Western part) becomes a rain shadow (dry) area.

31. Arrange the following seasons in Chronological order according to Hindu Calendar :

- (i) Autumn
- (ii) Summer
- (iii) Spring
- (iv) Rainy

Select the correct answer using the code given below:

- (a) (ii), (iv), (iii) and (i)
- (b) (iii), (ii), (iv) and (i)
- (c) (iv), (ii), (i) and (iii)
- (d) (i), (iv), (ii) and (iii)

U.P.R.O. / A.R.O. (Pre) 2021

Ans. (b)

According to Indian tradition, the year is divided into bimonthly segments in six seasons as follows :

Seasons	Months as per English Calendar
Spring	March & April
Summer	May & June
Rainy	July & August
Autumn	September & October
Pre-winter	November & December
Winter	January & February

32. Which one of the following statements is not correct?

- (a) The variability of rainfall at Chennai is more than that of Mumbai.
- (b) The retreating monsoon causes some rains in the coastal areas of Tamil Nadu.
- (c) The western disturbances originate from the Mediterranean Sea.
- (d) The Monsoon Expedition (MONEX) was organized under the joint auspices of the United States of America (U.S.A.) and India.

Raj. P.C.S. (Pre) 2024

Ans. (d)

To understand the causes and mechanism of the Monsoon, India and the USSR (Then the USSR, now Russia) conducted a joint Monsoon expedition, MONEX-1973, in the year 1973. It is noteworthy that in 1967, the World Meteorological Organization (WMO) and the International Council of Scientific Unions (ICSU) organized a Global Atmospheric Research Program (GARP). Under this program, the Global Weather Experiment began on December 1, 1978. A sub-program of this Global Weather Experiment was the Monsoon Experiment (MONEX-1979), which agreed to collaborate on certain scientific programs related to the experiment. The question is about the Monsoon Expedition (MONEX), not the Monsoon Experiment (MONEX). Therefore, option (d) is not correct.

33. Koppen has used which of the following words for the climate of the Great Plains of India?

- (a) Amw
- (b) As
- (c) Cwg
- (d) Aw

Uttarakhand P.C.S. (Pre) 2021

Ans. (c)

Cwg climate is often found in most parts of the Northern Ganga Plains. The summer temperature rises to approximately 40°C which falls to approximately 27°C in the winter. Significantly Koppen has used Cwg for the climate of Great Plains of India. Cwg type climate is also known as Monsoon type with dry winters.